OFFICIAL OPERATIONAL REFERENCE • VERSION 1.0

USER MANUAL

B2B CIRCULAR ECONOMY AUCTION ENGINE & BY-PRODUCT TRADE PLATFORM

Standard operating procedure and end-user workflow manual for industrial bulk buyers, secondary material generators, logistics coordinators, and state pollution control board (SPCB) compliance auditors.

12,500+ MT
INDUSTRIAL BY-PRODUCTS DIVERTED

₹8.2 Cr+
CIRCULAR VALUE GENERATED

150+ Heavy Plants
VERIFIED SPCB MANUFACTURING HUBS

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Live Deployment: https://trasure-phi.vercel.app

GitHub Repository: https://github.com/nivsorathiya-web/TRASURE

1. System Overview & Architectural Concept

Project TRASURE is an enterprise B2B circular economy platform developed to streamline the sourcing, freight logistics estimation, dynamic reverse auctioning, and statutory environmental compliance of industrial bulk by-products. Heavy process industries—such as thermal power stations, chemical manufacturing plants, fertilizer complexes, and metallurgical refineries—generate massive volumes of secondary materials (including Fly Ash, Ground Granulated Blast-Furnace Slag [GGBS], Spent Sulfuric Acid, and Phosphogypsum).

TRASURE provides industrial procurement managers and plant sustainability heads with a centralized, transparent web console to:

1. **Discover True Delivered Landed Economics:** Avoid misleading ex-factory rates by dynamically calculating delivered landed prices that factor actual highway transit distance, heavy freight tariffs (₹/km/MT), and terminal handling fees.
2. **Mitigate Road Freight Logistics Risks:** Real-time highway transit routing powered by Leaflet and OpenStreetMap/OSRM routing engines, computing exact driving polylines, highway distances, and estimated truck transit times across India.
3. **Execute Transparent Live Reverse Auctions:** Procurement buyers initiate time-bound reverse tenders where verified industrial sellers compete decremental bids in real time with anti-sniping protection.
4. **Automate Environmental Regulatory Compliance:** Instant generation of State Pollution Control Board (SPCB) Form 10 hazardous waste manifests, vehicle transport passes, Hazchem coding, and GST tax invoices.

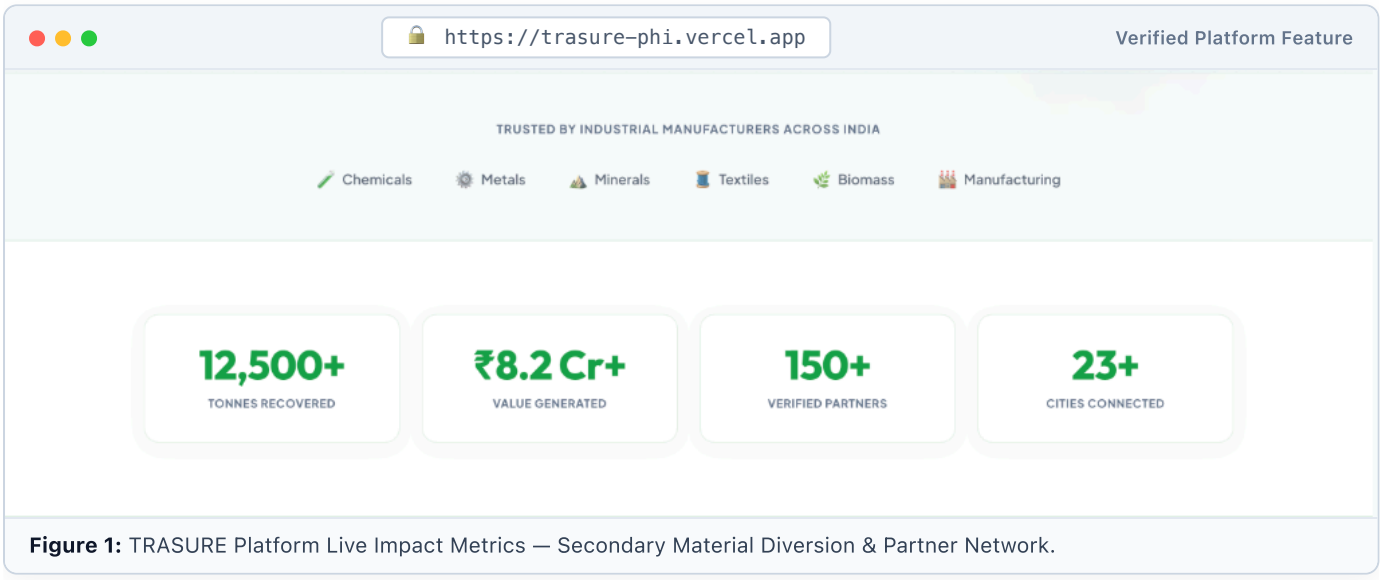


Figure 1: TRASURE Platform Live Impact Metrics — Secondary Material Diversion & Partner Network.



Figure 2: Three Steps to Circular Symbiosis — Onboard & Register, List or Search, and Compare & Trade.

2. System Requirements & Getting Started

Hardware & Software Prerequisites

COMPONENT	MINIMUM SPECIFICATION	RECOMMENDED SPECIFICATION
Web Browser	Google Chrome 90+, Mozilla Firefox 88+, MS Edge 90+, Safari 14+	Latest Chromium-based browser (Chrome / Edge / Brave)
Display Resolution	1280 × 720 (HD Ready Desktop)	1920 × 1080 (Full HD Desktop)
Network Connection	512 Kbps broadband (Initial asset load)	2 Mbps+ broadband (Fast map tile streaming)
JavaScript Engine	ES6+ compliant runtime (V8 / SpiderMonkey)	Enabled (Strict requirement for map rendering)
Local Storage	HTML5 LocalStorage enabled (5 MB allocation)	Enabled (Maintains plant profile, bids, invoices)

How to Launch the Application

1. **Via Production Cloud URL:** Open any modern web browser and navigate to: **https://trasure-phi.vercel.app**
2. **Via Local Workspace Clone:** If running from the local project repository:

```
git clone https://github.com/nivsorathiya-web/TRASURE.git
cd TRASURE
# Open index.html directly in your browser or serve via:
python3 -m http.server 8000
# Then visit http://localhost:8000
```

3. **Select Corporate Plant Profile:** On the login screen, select your registered plant (e.g., *Ambuja Cements Ltd — Ahmedabad, Gujarat*) and enter the Demo PIN 1234. Click **Access Corporate Console →** to enter the operational workspace.

3. System Navigation & Operational Layout

The TRASURE web interface is structured into three cohesive operational zones designed for maximum situational awareness and efficiency during procurement sessions:

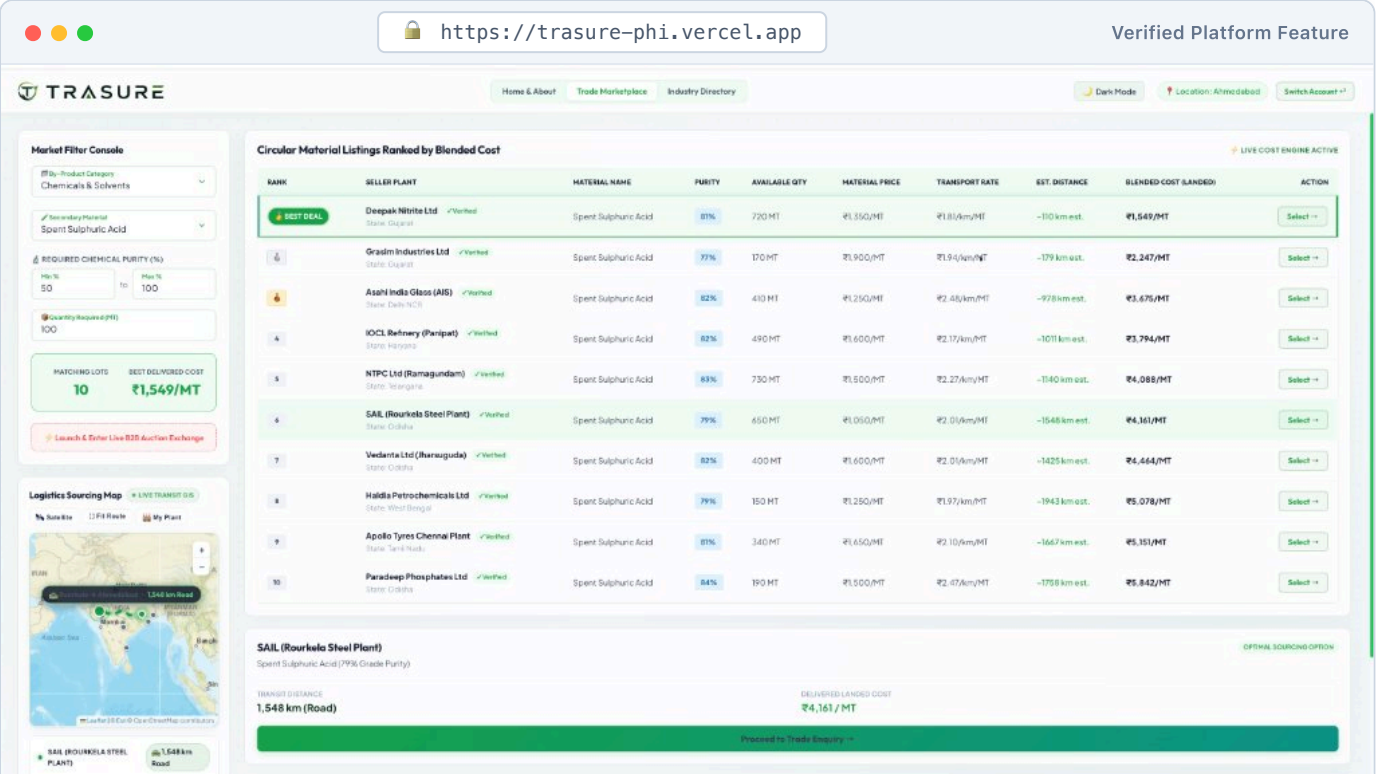


Figure 3: Project TRASURE Main Operational Console — Sourcing Filter Console, Ranked Circular Material Listings, and Real-Time GIS Logistics Route Map.

INTERFACE ZONE	COMPONENT ROLE	PRIMARY USER OPERATIONS & FEATURES
Top Navigation Bar	Global Status & Navigation	Displays platform brand identity, active plant location hub (*Ahmedabad*), Dark Mode toggle, and navigation tabs switching between **Home & About**, **Trade Marketplace**, and **Industry Directory**.
Zone 1: Market Filter Console (Left)	Multi-Parameter Sourcing Controls	Allows procurement buyers to filter by By-Product Category (*Chemicals & Solvents*), select secondary materials (*Spent Sulphuric Acid*), calibrate minimum required chemical purity (50% to 100%), and set required quantity in Metric Tonnes (MT). Also provides an instant launch button to enter the **Live B2B Reverse Auction Exchange**.
Zone 2: Ranked Material Listings (Center)	Delivered Landed Cost Engine	Displays a dynamic ranked leaderboard of verified supplier plant lots sorted by true delivered landed price. Each row details seller plant name, state, material purity percentage, available quantity (MT), ex-plant price, transport freight rate (₹/km/MT), driving distance, and the final all-inclusive delivered price per MT. Includes an instant **Select →** action button.
Zone 3: GIS Logistics Sourcing Map (Lower-Left / Drawer)	Leaflet Route Viewport & Summary	Renders an interactive map powered by Esri and Leaflet showing the selected highway transit corridor (e.g., *SAIL Rourkela to Ahmedabad: 1,548 km*), verified highway route badges, transit duration estimates (~28 hrs), and dynamic freight tariffs.

4. Core Features & Step-by-Step Procedures

Module 1: Material Sourcing Catalog & Landed Cost Discovery

Traditional industrial marketplaces only publish ex-factory prices, leading to severe logistical cost surprises when moving heavy bulk materials over hundreds of kilometers. TRASURE solves this by integrating the **Delivered Landed Cost Formula** directly into the search catalog:

```
Delivered Landed Cost (₹/MT) = Ex-Plant Base Price + (Road Distance in km × Freight Rate per km per MT) +
```

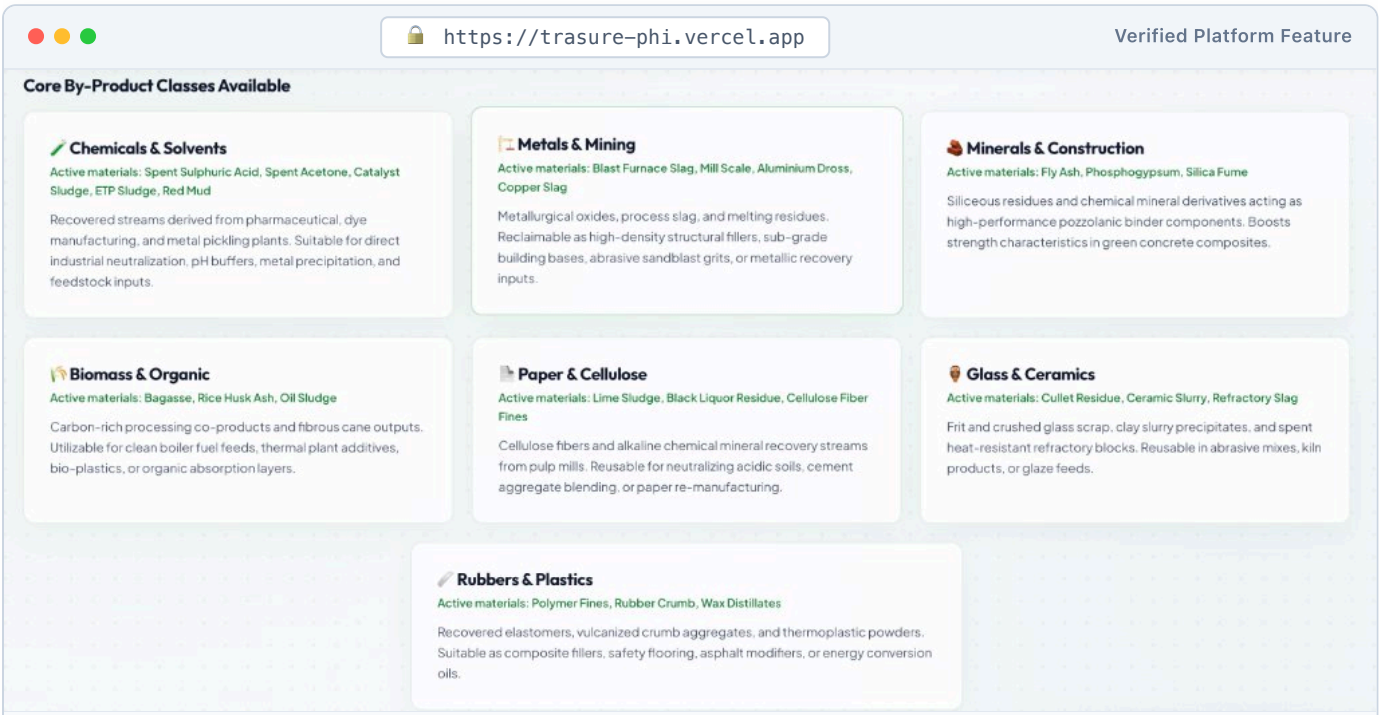


Figure 4: Core Industrial By-Product Classes Available on TRASURE — Chemicals, Metals, Minerals, Biomass, Paper, Glass, and Polymers.

Step-by-Step Procedure: Sourcing & Price Discovery

1. Navigate to the **Trade Marketplace** tab in the top navigation bar.
2. In the **Market Filter Console** on the left, select your target **By-Product Category** (e.g., *Chemicals & Solvents* or *Minerals & Construction*).
3. Select the exact secondary by-product material (e.g., *Spent Sulphuric Acid* or *Fly Ash Class F*).
4. Adjust the **Required Chemical Purity Slider** (e.g., 80% to 100%) and enter required lot tonnage (e.g., 100 MT).
5. The central **Circular Material Listings** table updates dynamically in real time, sorting available industrial plant supplies from lowest to highest Delivered Landed Cost. The optimal supplier is highlighted with a green **BEST DEAL** badge.
6. Click **Select →** on any supplier card to inspect its highway logistics corridor and proceed to trade enquiry or auction tender.

Module 2: GIS Logistics Transit Routing & Map Exploration

Project TRASURE couples industrial listings directly with precision GIS highway network routing powered by Leaflet and OpenStreetMap/OSRM APIs. It plots genuine driving routes along national highways rather than misleading straight-line distances.

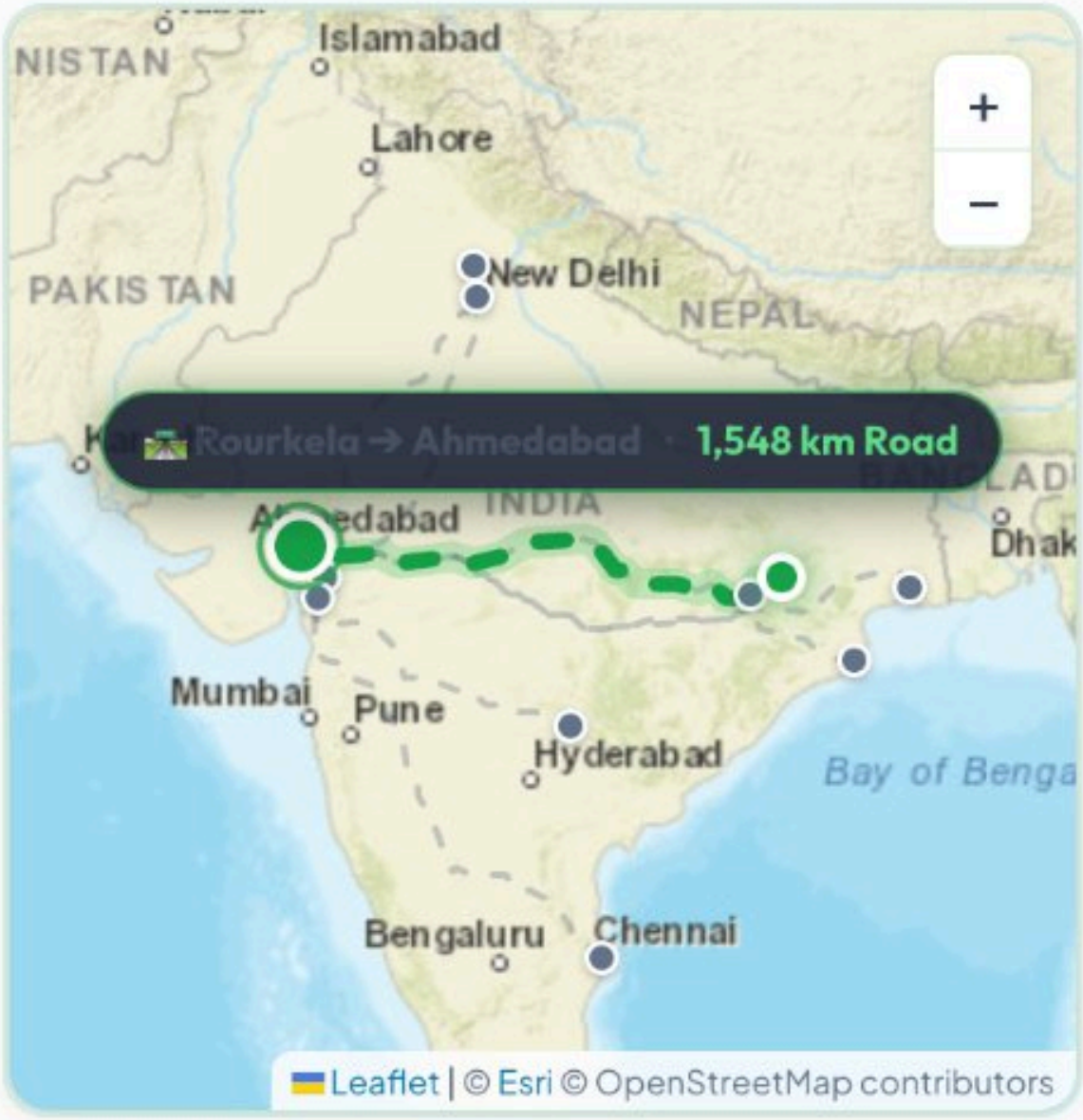
https://trasure-phi.vercel.app

Verified Platform Feature

Logistics Sourcing Map

LIVE TRANSIT GIS

Satellite | Fit Route | My Plant



SAIL (ROURKELA STEEL PLANT)

1,548 km Road

ORIGIN PLANT
Rourkela

→

DELIVERY HUB
Ahmedabad

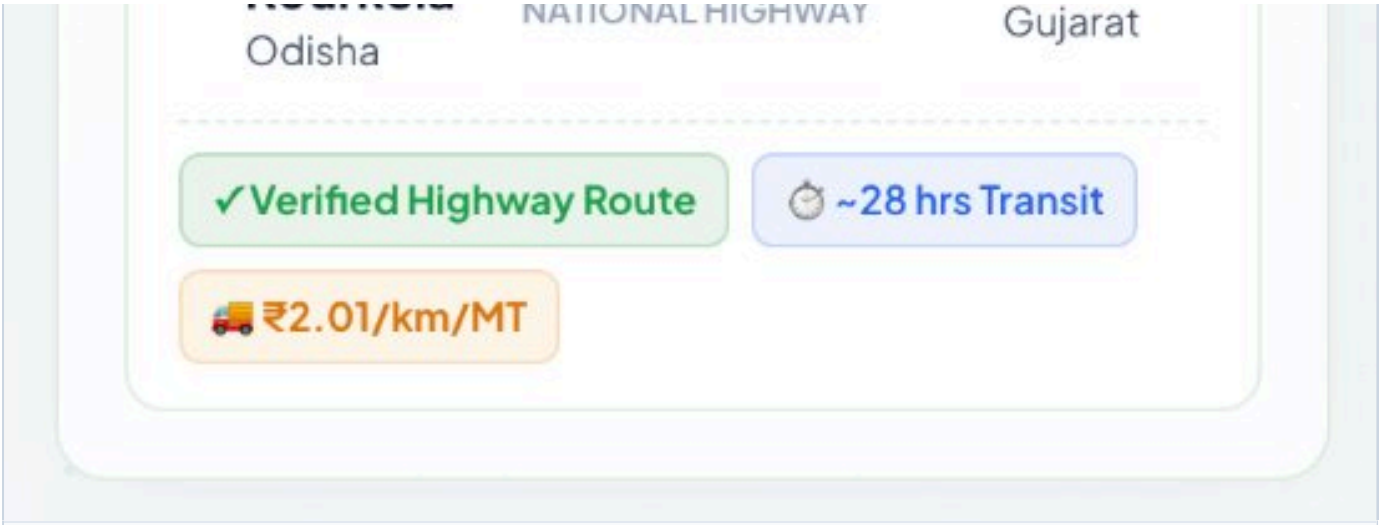


Figure 5: Logistics Sourcing Map — Real-Time Highway Transit Routing (Rourkela Steel Plant to Ahmedabad Hub, 1,548 km, NH Route, ~28 hrs transit).

Step-by-Step Procedure: Route Inspection

1. Select any supplier listing from the marketplace table or click a plant pin on the map.
2. The **Logistics Sourcing Map** viewport automatically pans and fits the viewport to the active route geometry.
3. The high-contrast highway polyline traces the exact route along national highways (e.g., NH-48 / NH-53).
4. Review the verified logistics briefing card:
    - **Origin Plant:** Plant name, industrial sector, and origin state.
    - **Delivery Hub:** Receiving plant location (e.g., *Ahmedabad, Gujarat*).
    - **Highway Distance:** Exact road transit distance in kilometers (e.g., *1,548 km Road*).
    - **Transit Duration:** Estimated transit time based on heavy commercial vehicle speeds (e.g., *~28 hrs Transit*).
    - **Freight Rate:** Dynamic haulage tariff per kilometer per metric tonne (e.g., *₹2.01/km/MT*).
5. Use the **Satellite** toggle to switch basemap imagery or click **Fit Route** to recalibrate zoom levels.

Module 3: Live B2B Reverse Auction Exchange & Bidding Desk

The reverse auction module allows industrial procurement managers to publish buying tenders where pre-qualified secondary material producers compete decremental bids to secure off-take volume.

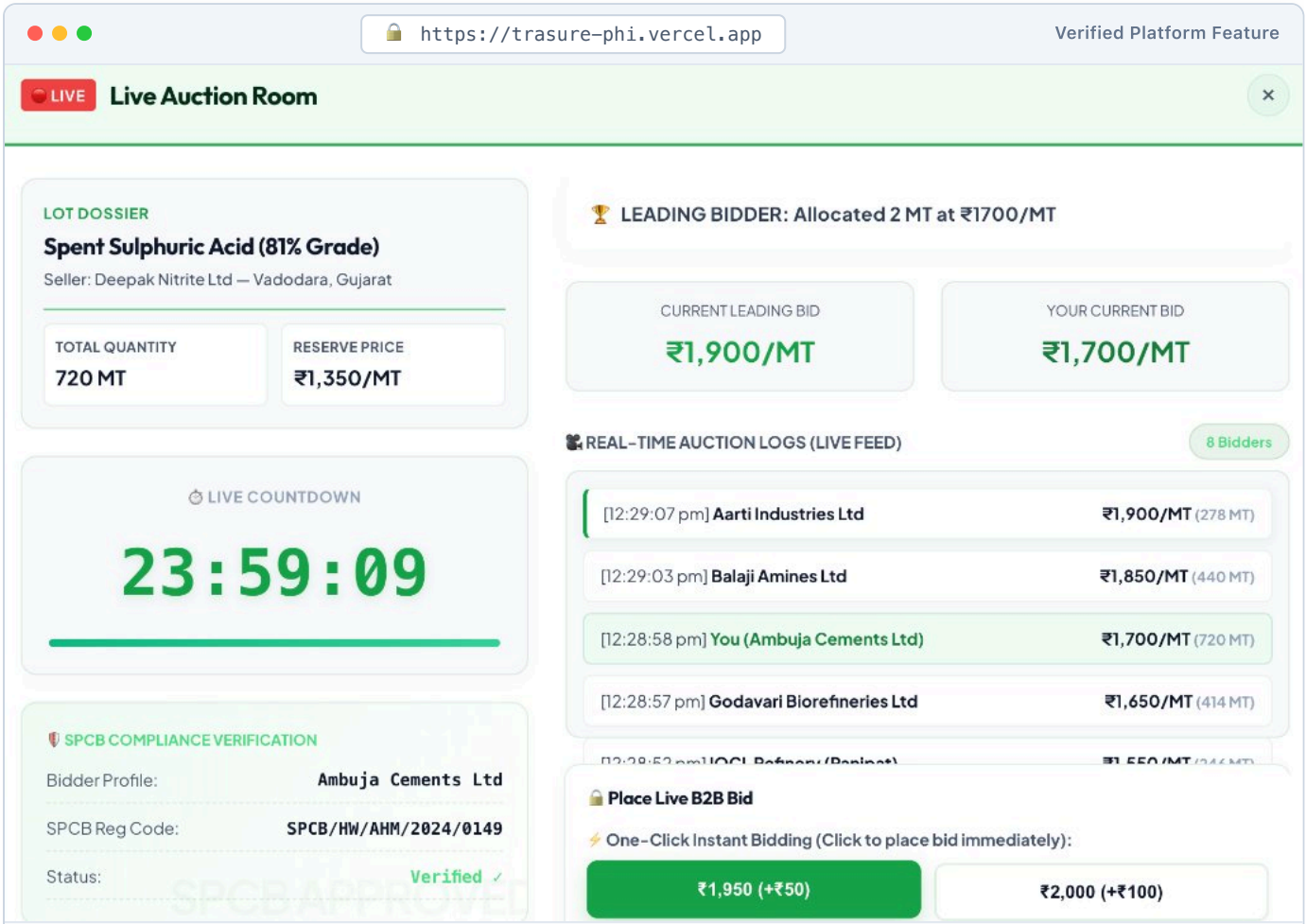


Figure 6: Live B2B Auction Room — Real-Time Countdown Timer (23:59:09), Verified SPCB Reg Code, Live Feed of Competitor Bids, and One-Click Decrement Bidding.

Auction Operational Rules:

- **Reverse Pricing Model:** The current lowest bid is the leading bid. Suppliers must submit decremental bids strictly below the current leading price.
- **Minimum Decrement Rule:** Every counter-bid must undercut the current leading rate by at least **₹50/MT** or **₹100/MT** to prevent trivial spam bidding.
- **Anti-Sniping Overtime Protection:** If any counter-bid is submitted when the live countdown timer has **less than 60 seconds remaining**, the system automatically triggers an anti-sniping extension, adding **+60 seconds** to the auction clock. This prevents predatory last-second sniping and guarantees fair price discovery.
- **Statutory SPCB Clearances:** Bidding is strictly restricted to enterprises with verified SPCB authorization codes (e.g., `SPCB/HW/AHM/2024/0149`).

Step-by-Step Procedure: Reverse Auctioning

1. Click ⚡ **Launch & Enter Live B2B Auction Exchange** from the Marketplace Console.
2. The **Live Auction Room modal** opens, displaying the Lot Dossier (Material: *Spent Sulphuric Acid 81% Grade*, Total Qty: *720 MT*, Reserve Ceiling Price: *₹1,350/MT*).
3. Monitor the **Live Countdown Timer** and the **Real-Time Auction Logs feed** displaying competitor bids from verified industrial plants (e.g., *Aarti Industries, Balaji Amines, Godavari Biorefineries*).
4. To place a bid, use the **One-Click Instant Bidding buttons**: click **₹1,950 (+₹50)** or **₹2,000 (+₹100)**.

5. Upon auction expiration (00:00:00), the auction floor freezes, the lowest bidder is declared the winner, and the settlement desk automatically opens to generate the SPCB Form 10 manifest and tax invoice.

Module 4: SPCB Environmental Compliance & Manifest Desk

Secondary industrial by-products (especially spent acids, metal slags, and chemical catalysts) fall under the regulatory scrutiny of State Pollution Control Boards (SPCB) and Central Pollution Control Board (CPCB) Hazardous and Other Wastes Rules. TRASURE ensures zero non-compliant off-take by enforcing automated regulatory gating.

Compliance Workflow & Form 10 Generation:

1. **Hazardous Material Classification:** The system automatically assigns statutory Hazchem classification codes (e.g., *Hazchem 2WE, UN 1830 Class 8 Corrosive* for Spent Sulphuric Acid).
2. **Authorized Consignment Numbering:** Generates unique statutory serial numbers in compliance with SPCB Form 10 guidelines (e.g., `TRS–HW–2026–0842`).
3. **Carrier & Driver Authorization:** Validates GPS tracking transponders, hazardous vehicle fitness certificates, and authorized driver licensing prior to dispatch clearance.

Module 5: Automated Invoicing & Financial Settlement Desk

Following trade conclusion, TRASURE generates commercial settlement tax invoices that decouple ex-plant material charges from transport freight charges for transparent GST accounting:

- **Material Base Subtotal:** Awarded Rate × Tonnage (e.g., ₹1,350 × 100 MT = ₹1,35,000).
- **Logistics Freight Subtotal:** Distance × Freight Tariff × Tonnage (e.g., 110 km × ₹1.81/km/MT × 100 MT = ₹19,910).
- **Terminal Handling Fee:** Flat handling and weighbridge charge (₹150 × 100 MT = ₹15,000).
- **Statutory GST:** 18% GST (split equally into 9% CGST and 9% SGST).
- **Printable & Audit-Ready:** Generates printable commercial invoices for corporate ERP record-keeping.

Module 6: System Administration & Plant Directory

Platform administrators and enterprise users can verify active industrial participants, inspect GSTIN registrations, and audit regional manufacturing capacity across India.

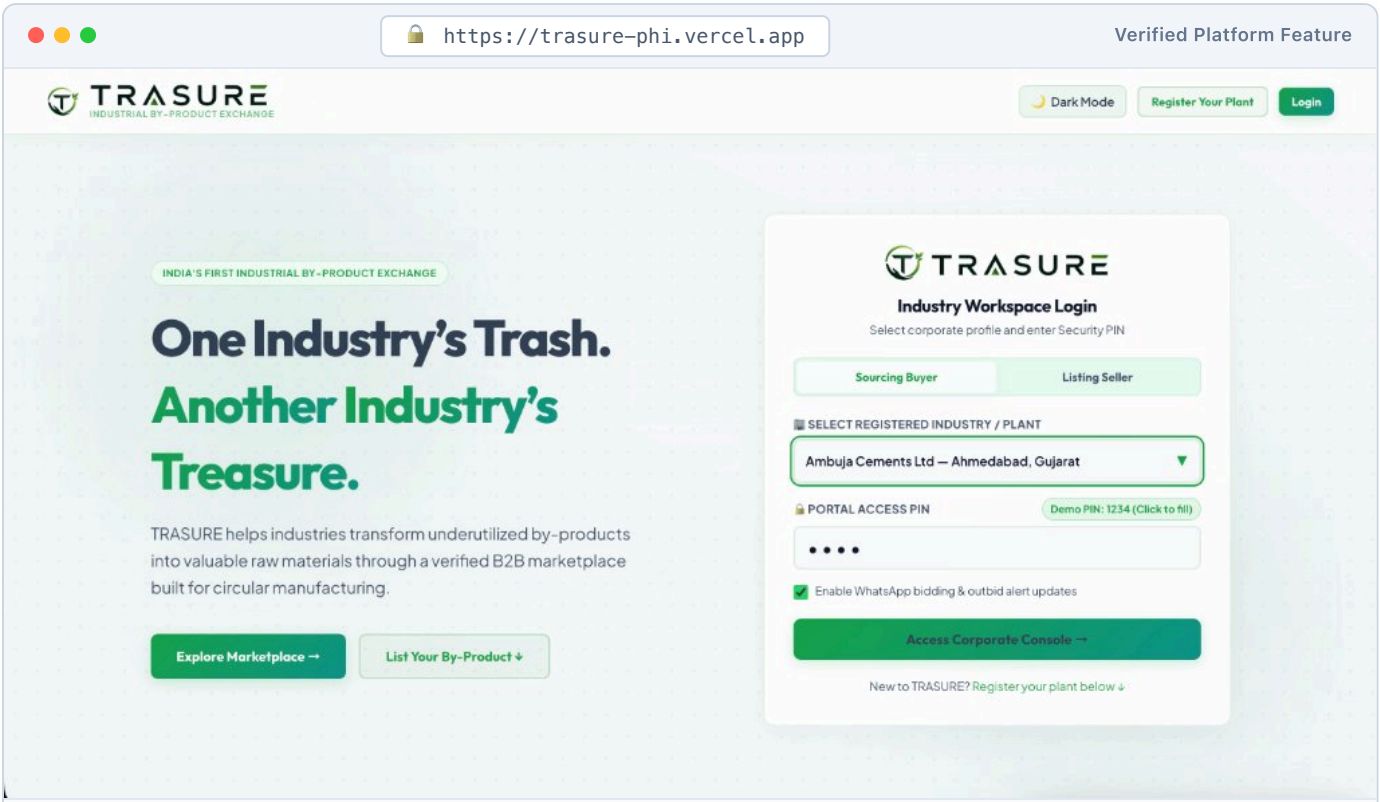


Figure 7: Homepage Hero & Industry Workspace Authentication Console — Single-Click Demo PIN (1234) and Registered Plant Selection.

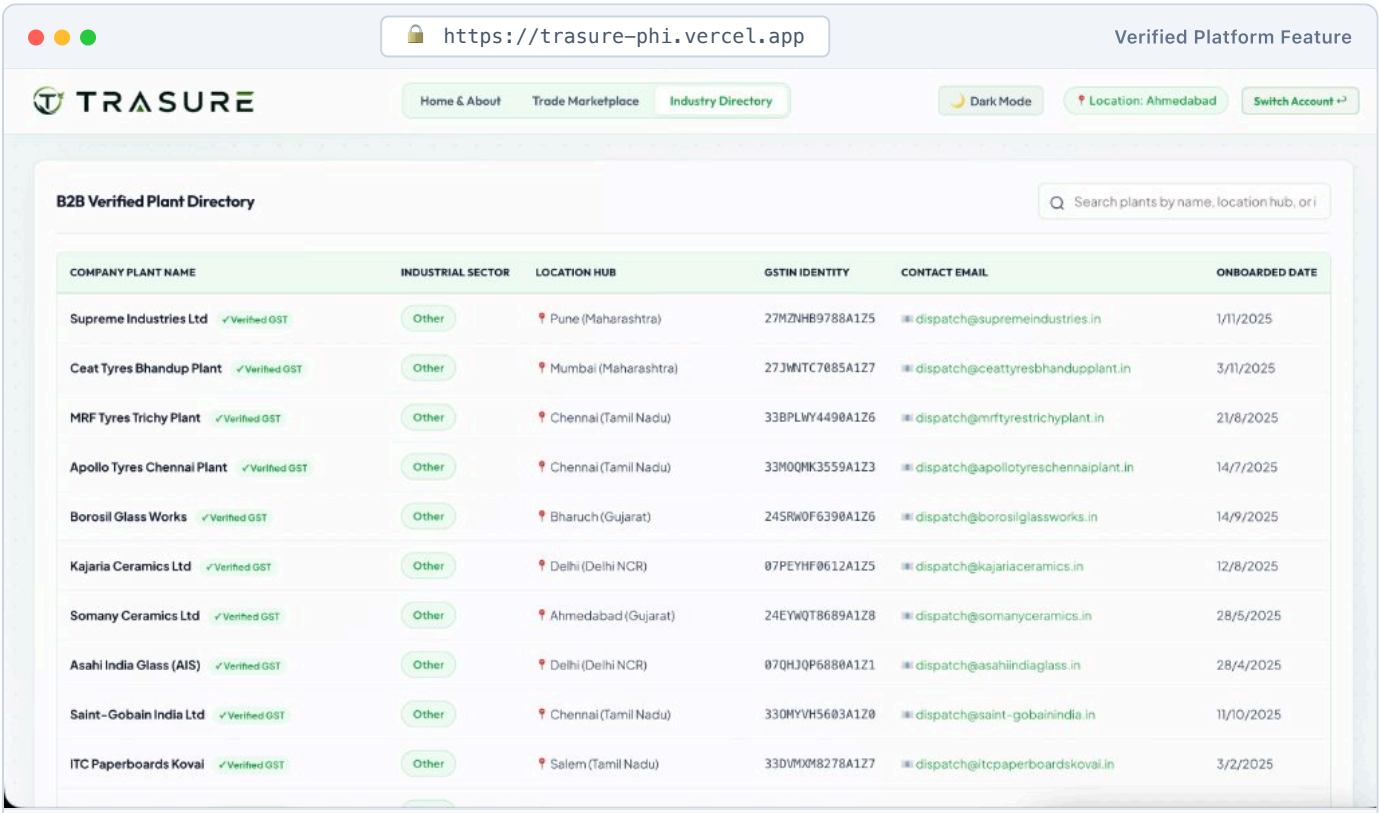


Figure 8: B2B Verified Plant Directory — Enterprise registry with verified GSTIN identities, industrial sectors, and location hubs.

5. Troubleshooting & Frequently Asked Questions

Frequently Asked Questions

QUESTION	RESOLUTION / ANSWER
Why does delivered cost differ from ex-plant price?	Delivered landed cost incorporates actual highway road freight (distance × tariff) and terminal handling fees, showing what the material actually costs upon arrival at your factory gate.
How does anti-sniping work in the reverse auction?	If any participant places a bid with less than 60 seconds left on the clock, the system automatically adds +60 seconds of overtime, ensuring all bidders have adequate time to respond.
How do I change my receiving factory destination?	Use the **Switch Account / Plant Selector** dropdown in the top header to choose your factory location (e.g., *Ahmedabad, Vadodara, Surat*). All distances and freight rates will immediately recalculate relative to that hub.
Are the highway distances calculated in straight lines?	No. TRASURE queries the OSRM road routing engine via real national highway corridors (NH-48, NH-53, etc.), providing true road kilometers and realistic driving times.

6. Technical Support & Verification Contacts

For software verification, academic evaluation, or technical assistance regarding Project TRASURE:

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